

PRATEEK TANWAR

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SUMMARY

Automation focused Backend Engineer specializing in production grade Python automation, RESTful API development, and enterprise scale data processing. Delivered multiple automation scripts and DRF microservices that eliminated manual workflows and improved operational reliability across Bharti Airtel's network operations. Strong foundation in backend development with expertise in Python, Django REST Framework, Database integration, Docker Containerization, and cross functional project delivery.

WORK EXPERIENCE

Backend Engineer - Bharti Airtel Ltd.

Jul 2024 – Present

- **Engineered 40+** production-grade Python automation scripts leveraging Netmiko and multithreading to streamline router upgrades, configurations, BRAS ticketing etc, **saving approximately 2.5 years of cumulative manual effort** while **reducing human error by 99%**.
- **Architected the complete backend for the ART Portal** using Django REST Framework, MySQL, and LDAP authentication, enabling comprehensive **request lifecycle tracking, developer performance dashboards**, and **automated Excel exports**, driving process transparency for Manesar and Chennai NOCs.
- **Developed a unified MIS Report Portal** integrating Django REST Framework with Oracle Database, providing users with **real-time report preview, download capabilities, and direct email dispatch**, centralizing NOC reporting and significantly reducing tool fragmentation.
- **Automated over 20 Excel-based MIS reports** through Python and Pandas, implementing auto-scheduled execution and emailing to eliminate manual report preparation and ensure timely, SLA compliant daily, weekly, and monthly report delivery.
- **Built ACE (Airtel Code Executor)**, a secure Django-based GUI tool **empowering non-technical users to safely execute automation scripts**, while enforcing container security and maintaining strict operational governance.
- **Led cross-functional collaboration with NOC teams** to gather requirements, author business requirement documents (BRDs), conduct risk assessments, and deliver audit compliant automation frameworks that minimize service outage risks.

GET Intern - Bharti Airtel Ltd.

Feb 2024 – Jun 2024

- **Analyzed and classified network alarms from from all Airtel OEMs** (Cisco, Huawei, Nokia, Juniper, ECI, Ciena) into **Service Affecting (SA)** and **Non-Service Affecting (NSA)** categories, enhancing incident prioritization and supporting Infra and Service Experience teams in efficient fault management.
- **Conducted root cause analysis (RCA)** of network issues to accelerate problem identification and resolution, contributing to improved network reliability and optimized operational performance.
- **Coordinated closely with IBM to develop a new service alarms dashboard**, enabling more effective monitoring and faster issue resolution.

[Link](#)

IT Intern - Hero MotoCorp Ltd.

Jul 2023 – Aug 2023

- **Gained comprehensive exposure to the "Smart Eye" project**, a **supervised machine learning** based vision system developed for **quality inspection in bike manufacturing**.
- Studied the system's use of six cameras mounted over a conveyor belt to capture multi-angle images of bike parts. The system processes these images to generate pass/fail classifications, verifying correct parts and ensuring only validated components proceed to assembly. This automation **improved verification accuracy and efficiency across manufacturing plants**, significantly **reducing human errors related to wrong part usage in export models**.

[Link](#)

EDUCATION

B.Tech in Computer Science Engineering

2020-2024

Maharshi Dayanand University, Rohtak

- Specialization in Artificial Intelligence and Machine Learning
- GPA:- 8.5/10

Higher Secondary (C.B.S.E) ,(PCM, CS)

2019-2020

Blue Bells Public School, Gurugram

- Percentage:- 96%/100

SKILLS

- **Languages:** Python, C++, JavaScript, HTML, CSS
- **Frameworks:** Django, Django REST Framework, FastAPI, React.js, Tailwind CSS
- **Database:** MySQL, Oracle DB
- **Systems & Networking:** Computer Networks, NOC Operations, Networking protocols, SSH, Telnet
- **DevOps & Tools:** Docker Containerization, Postman, Git, Linux, Netmiko
- **Concepts & Methodologies:** REST APIs, Microservices, Multithreading, SDLC
- **Collaboration:** Cross-functional Teamwork, Project Delivery

PROJECTS

- **AutoReq Pro:** Developed a Jira-inspired internal platform for submitting, tracking, and managing project requests, featuring role-based dashboards, real-time status updates, automated email notifications, testing coordination, developer performance tracking, and Excel export capabilities. This platform significantly improves efficiency and transparency across teams.
Tech Stack: Django REST Framework (Backend), React.js (Frontend), MySQL (Database)
- **Driver's Drowsiness Detection System Using Machine Learning:** Designed and implemented a real-time driver drowsiness detection system utilizing machine learning techniques to enhance road safety by preventing accidents related to drowsy driving.
Tech Stack: Python, OpenCV, dlib, MySQL, Machine Learning [Link](#)
- **CloudChaser Weather React App:** Built a React.js application that fetches and displays real-time weather data using the OpenWeatherMap API, providing accurate weather information worldwide.
Tech Stack: HTML, CSS, JavaScript, React.js, Netlify, OpenWeatherMap API [Link](#)
- **WhatsApp Chat Portfolio:** Created a web based chat application with an interface similar to WhatsApp, serving as an interactive portfolio showcase.
Tech Stack: HTML, CSS, JavaScript [Link](#)

ADDITIONAL INFORMATION

- **Certifications:** C++ Programming By Coding Ninjas, Data Structures & Algorithms By Coding Ninjas, Python Programming By IIT Bombay, SQL By HackerRank, Machine Learning By Coursera
- **Awards/Activities:** Recipient of the **Airtel ACE Award** in recognition of outstanding effort and valuable contributions to the organization [Link](#)
- **Languages:** English, Hindi